Latent Action Pretraining from Videos

Pretraining a robot policy on video, with no action labels

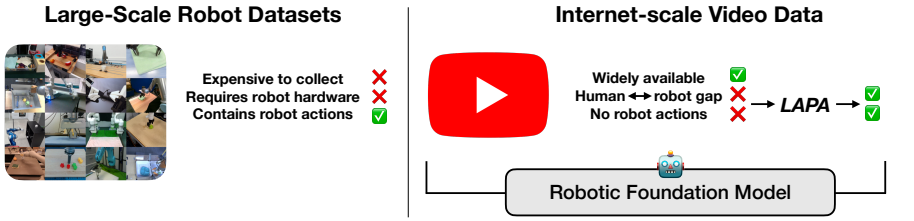
Juntang Wang · Vivian Hui

Harvard CS 2883R · 18 Nov 2026

Seonghyeon Ye, Joel Jang, et al.: KAIST, UW, Microsoft Research, NVIDIA, AI2.

Latent Action Pretraining from Videos. ICLR 2025. arXiv 2410.11758

Problem



Ye et al. (2025), Fig. 1

- A **VLA** maps an image and an instruction to a robot action (Brohan et al., 2023; Kim et al., 2024)
- Its training data is teleoperated: OpenVLA was pretrained on 970k demos, recorded one at a time
- Web video is free and endless. Can it do the pretraining instead, with no action labels?

Before LAPA

VPT

(Baker et al., 2022)

- Train an inverse dynamics model on labelled data, then let it label the video
- Still needs labels, in the target robot's action space

UniPi

(Du et al., 2023)

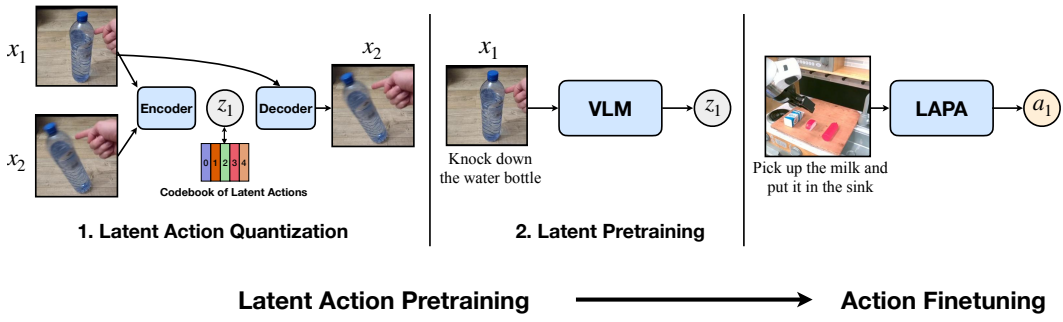
- Generate the future frames, then read actions off them
- No labels to pretrain, but the generated plans fail on long tasks

Latent actions

(Bruce et al., 2024; Schmidt and Jiang, 2024)

- Genie and LAPO learn actions from frames alone, in games
- LAPA takes this to real video and a 7B model

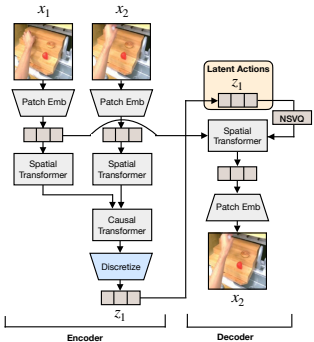
Method



Ye et al. (2025), Fig. 2

- Learn a vocabulary of visual changes, learn when to use each one, then translate it into robot commands
- Stages 1 and 2 see the same unlabelled video. Only stage 3 sees a robot action.

Stage 1: a vocabulary of visual changes



Ye et al. (2025), Fig. 8

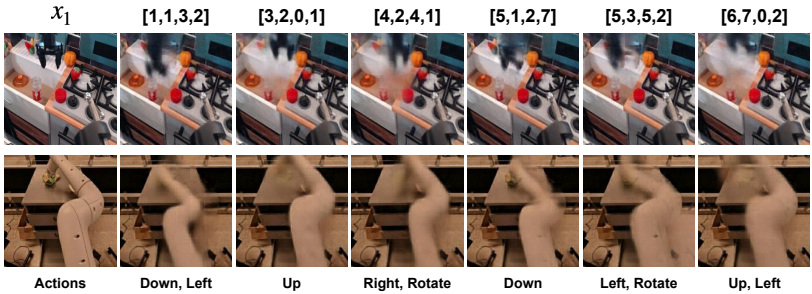
$$d_1 = e_2 - e_1, \qquad z_1 = \arg\min_{z_k \in C} \|d_1 - z_k\|_2$$

$$\mathcal{L} = \|x_2 - D(x_1, z_1)\|_2^2$$

e_i: frame embedding · C: codebook · D: decoder · full objective in the backup

- The code is the *difference* between two frames 0.6 s apart, snapped to the nearest of 4,096 entries
- The decoder is given x_1 already, so the only thing left for the code to carry is what changed
- Encoder reads as an inverse dynamics model, decoder as a world model

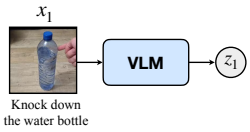
What a code looks like



Ye et al. (2025), Fig. 6

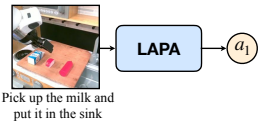
- Each column is one code, decoded into two different scenes and robot bodies
- The same code gives the same coarse motion in both rows, so the vocabulary is shared across bodies
- Nothing forces this: in human video, some codes turn out to be camera motion (backup)

Stages 2 and 3: pretrain on codes, fine-tune to actions



2. Latent pretraining

- Image + instruction → code, supervised by stage 1's encoder
- A 7B video-language model (Liu et al., 2024), a new head over the codes, one epoch
- Any video with a language caption. No robot.

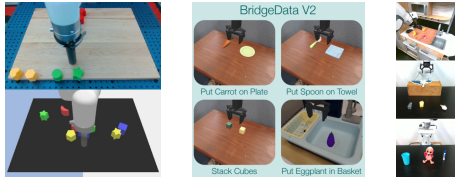


3. Action fine-tuning

- Throw the code head away; a fresh head predicts real actions
- Actions are small changes to the gripper: move, rotate, open or close
- 100 to 7k labelled demos, against OpenVLA's 970k

Ye et al. (2025), Fig. 2, middle and right

Setup



Ye et al. (2025), Fig. 9: Language Table and SIMPLER, both simulated, and a real Franka arm

- Pretrain with the action labels withheld, on simulated video, Bridge (one robot), Open-X (many robots), or human video
- Fine-tune on 100 to 7k labelled demos of the test robot, then test on seen and unseen objects and instructions
- The question each time: does pretraining without labels beat pretraining with them?

Bridge (Walke et al., 2023), Open-X (Open X-Embodiment Collaboration, 2023), human video (Goyal et al., 2017). All eight settings in the backup.

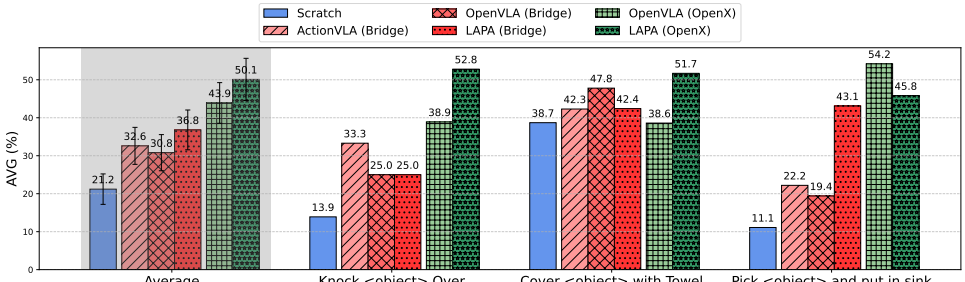
Results: Language Table

	in-domain (1k)		cross-task (7k)		cross-env (1k)	
	seen	unseen	seen	unseen	seen	unseen
Scratch	15.6±9.2	15.2±8.3	27.2±13.6	22.4±11.0	15.6±9.2	15.2±8.3
UniPi	22.0±12.5	13.2±7.7	20.8±12.0	16.0±9.1	13.6±8.6	12.0±7.5
VPT	44.0±7.5	32.8±4.6	72.0±6.8	**60.8**±6.6	18.0±7.7	18.4±9.7
LAPA	**62.0**±8.7	**49.6**±9.5	**73.2**±6.8	54.8±9.1	**33.6**±12.7	**29.6**±12.0
ActionVLA	77.0±3.5	58.8±6.6	77.0±3.5	58.8±6.6	64.8±5.2	54.0±7.0

Ye et al. (2025), Tab. 1: success rate (%) ± std. err.; bold = best of the top four rows; ActionVLA pretrains on the true actions

- LAPA leads 5 of 6 columns. VPT wins cross-task unseen, where its labeller was trained on the 7k labelled demos
- Cross-env, pretrained on real video and tested in simulation, is where a label-trained labeller stops transferring and LAPA still helps
- The error bars overlap almost everywhere: read the ordering, not the gaps

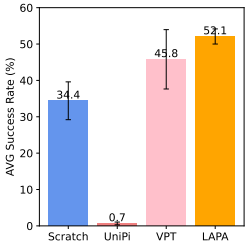
Results: real robot

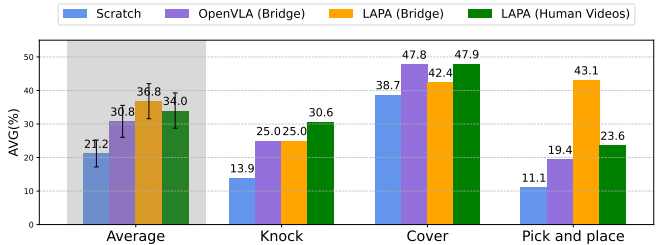


Ye et al. (2025), Fig. 3

- Franka arm, 3 tasks, 54 attempts each, partial credit for reaching and grasping
- Pretrained on Open-X: LAPA 50.1 against OpenVLA's 43.9, ahead on all three kinds of generalization
- Pretrained on Bridge, LAPA also beats both action-labelled models. The authors' reading: labels overfit the policy to one robot's action space

Results: human video only





Ye et al. (2025), Fig. 4: (a) SIMPLER, (b) real robot

- Pretrained on Something-Something v2 alone: no robot, no actions, a 2.4 s window
- Real robot 34.0: above OpenVLA pretrained on Bridge (30.8), below OpenVLA pretrained on Open-X (43.9)
- The three test tasks suit the corpus: it holds 6,655 “knocking” clips where Bridge holds 2

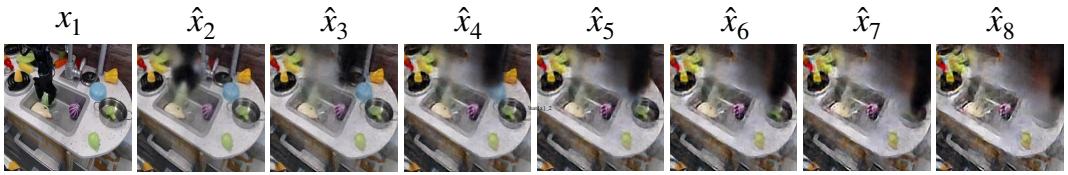
Pretraining cost

	LAPA (Open-X)	OpenVLA
GPU-hours	272 H100	21,500 A100
batch size	128	2048
output space per step	8^4	256^7

Ye et al. (2025), §4.6: "approximately 30–40 times more efficient"

- The authors attribute it to the backbone, already trained on video, and to the tiny output space
- Not an end-to-end audit: two different accelerators, and the cost of training stage 1 is not reported

Policy plus decoder



Ye et al. (2025), Fig. 7

- A pretrained LAPA, never fine-tuned, picks codes for “take the broccoli out of the pot”
- Stage 1’s decoder renders each next frame, and that frame becomes the next input: a closed loop with no robot
- Eight frames, one example, no metric. Suggestive of a world model, not yet evidence of one

Limitations

Stated (§6)

- A code spans 0.6 s, too coarse for grasping
- 7B inference latency, as for any VLA
- Manipulation only; no driving or navigation

Ours

- The language labels are not free. All three corpora ship with text
- LAPA and OpenVLA tie in 51.9% of the real-robot attempts
- Codes absorb camera motion, and the policy sees one frame, no history
- The SIMPLER fine-tuning demos came from a Bridge-trained policy

Takeaway

- LAPA replaces the action labels *in pretraining* with a learned vocabulary of visual changes. Downstream it still needs demos, just a hundred or so instead of a million
- Where it wins is crossing from one robot body to another, which is exactly where action labels turn out to hurt
- What is shown is strong; what is claimed around it, a world model and label-free robotics, is not shown yet

Discussion

Roles, in call order

From the Ed post and the role sheet (second tab). 5 minutes each.

Open questions

- What can a code between two frames never capture: force, grasp timing, speed?
- Is the gain from the latent actions, or from a backbone already trained on video? Nothing in the paper swaps it out
- What would have to be true before a decoder rollout could stand in for a real-robot trial?

Backup: stage 1 objective

$$z_1 = \arg\min_{z_k \in C} \|d_1 - z_k\|_2, \quad d_1 = e_2 - e_1 \tag{1}$$

$$\hat{d}_1 = d_1 + \frac{\|d_1 - z_1\|}{\|v\|} v, \quad v \sim \mathcal{N}(0, 1) \tag{2}$$

$$\hat{x}_2 = D\big(\mathrm{Attn}(\mathrm{sg}[p_1], \hat{d}_1, \hat{d}_1)\big) \tag{3}$$

$$\mathcal{L} = \|x_2 - \hat{x}_2\|_2^2 \tag{4}$$

Ye et al. (2025), App. A

- e_1, e_2: C-ViViT (Villegas et al., 2023) embeddings of x_1 and x_2; p_1: patch embedding of x_1; C: the codebook
- (2) is NSVQ (Vali and Bäckström, 2022): substitute noise of the same magnitude as the quantization error, so the codebook stays differentiable and codes stop dying
- (3) stops the gradient through p_1, or the encoder and decoder collude and the code carries nothing

Backup: the eight experiments

benchmark	test	pretrain on, no labels	trajs	fine-tune
Language Table (Lynch et al., 2023)	in-domain	sim	181k	1k
	cross-task	sim	181k	7k
	cross-env	real	442k	1k
SIMPLER (Li et al., 2024)	in-domain	Bridge	60k	100
	cross-body	human video	200k	100
Real Franka	cross-body	Bridge	60k	450
	many bodies	Open-X	970k	450
	cross-body	human video	200k	450

Ye et al. (2025), Tab. 3

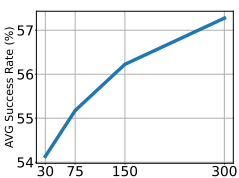
Bridge (Walke et al., 2023); Open-X (Open X-Embodiment Collaboration, 2023); human video: Something-Something v2 (Goyal et al., 2017). Tab. 3 gives 200k Something-Something videos, §4.5 gives 220K.

Backup: real robot by generalization type

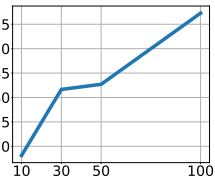
	seen objects, unseen combo	unseen objects	seen objects, unseen instr.	average
Scratch	18.0	20.3	25.4	21.2
ActionVLA (Bridge)	38.3	31.8	27.7	32.6
OpenVLA (Bridge)	35.6	34.6	22.1	30.8
LAPA (Bridge)	43.4	31.4	35.6	36.8
OpenVLA (Open-X)	46.2	42.1	43.4	43.9
LAPA (Open-X)	**57.8**	**43.9**	**48.5**	**50.1**
LAPA (human video)	36.5	37.4	28.1	34.0

Ye et al. (2025), Tab. 2: success rate (%), partial credit, 54 attempts per model

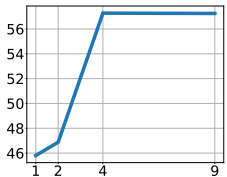
Backup: scaling



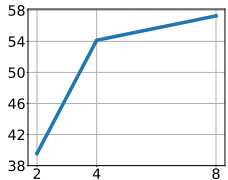
model size (M)



data (% of Bridge)



latent length

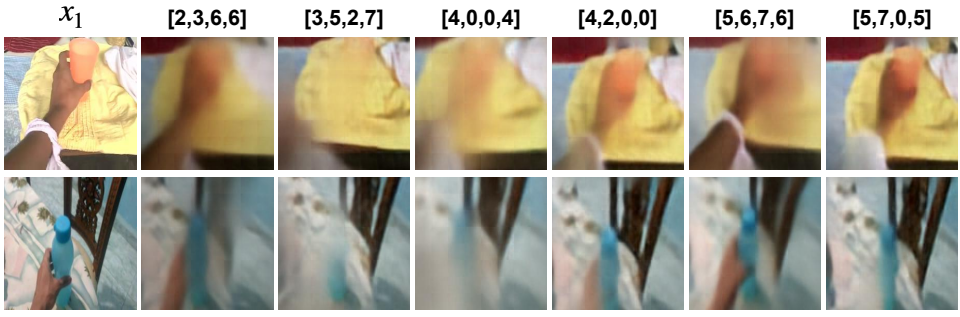


latent vocabulary

Ye et al. (2025), Fig. 5: average success rate (%)

§5.1 reads panel (c) as Language Table; App. F describes Fig. 5 as Bridge and SIMPLER throughout. Check before quoting a single panel.

Backup: camera motion as a code



Ye et al. (2025), Fig. 14

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